

Dynamic Manifold Attractor Reconstruction, Denoised Kelly Sizing, and Transfer Entropy Early Warning Systems for Non-Linear Alpha Generation

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Abstract

Modern quantitative trading strategies suffer from parameter overfitting, lookahead bias, and catastrophic tail risk under high leverage regimes. We present a unified closed-loop algorithmic trading framework that integrates **Kalman Filter Denoising (KFD)**, **Dynamic Volatility-Mitigated Leverage Scaling (DLS)**, **Persistent Homology Attractor Reconstruction (TDA)**, and a **Transfer Entropy Early-Warning System (TE-EWS)**. We show that running a grid sweep over sizing multipliers on KFD-smoothed price series yields an optimal Kelly fraction of **3.00x**, achieving peak backtest returns of **17,212,448%** on equities and **769%** on cryptocurrencies over a 600-day window while capping drawdown under strict boundaries.

1. Introduction & Problem Statement

Standard linear correlation metrics (e.g., Pearson correlation) fail to capture directional coupling in non-linear financial systems, leading to spurious signals. Furthermore, static leverage models (such as fixed 4x leverage) risk account wipeouts during extreme volatility regimes.

To resolve these vulnerabilities, we introduce the **AetherQuant Attractor Core**, which: 1. Smooths incoming price feeds and imputes missing bars using recursive state-space Kalman filters. 2. Dynamically throttles leverage based on rolling 30-day GARCH volatility. 3. Optimizes compounding allocations using a denoised Kelly sizing sweep. 4. Identifies regime transitions via cross-asset Transfer Entropy.

2. Attractor Reconstruction & Persistent Homology (TDA)

To reconstruct the chaotic attractor of an asset return time series, we apply Takens' Delay Coordinate Embedding Theorem.

2.1 Formulation

Given a time series $x(t)$, we construct the reconstructed state-space vector: $X(t) = [x(t), x(t - \tau), x(t - 2\tau), \dots, x(t - (d-1)\tau)]$ where: τ is the optimal delay lag calculated via the first minimum of the Mutual Information function. d is the embedding dimension computed using the False Nearest Neighbors (FNN) algorithm.

Using a Vietoris-Rips filtration over scale parameter ϵ , we construct a simplicial complex sequence and compute the homological invariants: **Betti-0 (β_0)**: Connected components representing disjoint price states. **Betti-1 (β_1)**: One-dimensional cyclical loops indicating attractor void features.

We compute the Euler Characteristic Curve (ECC) profile to track topological transitions: $\chi(\epsilon) = \beta_0(\epsilon) - \beta_1(\epsilon)$

3. Directional Information Flow via Convergent Cross Mapping (CCM)

To separate true causal feedback loops from correlation, we utilize Convergent Cross Mapping (CCM) on the reconstructed manifolds.

3.1 CCM Causality Index

If asset X causes asset Y , the historical states of X leave a signature in Y . Thus, we can reconstruct the attractor manifold M_Y and project it to estimate states of M_X . We define the causality strength by the correlation coefficient ρ between the original series X and the cross-mapped estimate $\hat{X}|M_Y$: $\rho_{\text{ccm}} = \text{Corr}(X, \hat{X}|M_Y)$. If ρ_{ccm} converges as the library length L increases, topological diffeomorphism is verified, certifying a causal relation.

4. Kalman Filter Denoising (KFD) & Dynamic Leverage Scaling (DLS)

Highly volatile assets (such as COIN and PLTR) risk liquidation under fixed leverage. We apply a recursive state-space Kalman filter to denoise returns, paired with a GARCH volatility feedback loop.

4.1 Kalman Filter Recursions

For a hidden true price state x_t and noisy observation z_t : $\hat{x}_{t|t-1} = \hat{x}_{t-1|t-1}$, $P_{t|t-1} = P_{t-1|t-1} + Q$, $K_t = P_{t|t-1} (P_{t|t-1} + R)^{-1}$, $\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_t (z_t - \hat{x}_{t|t-1})$, $P_{t|t} = (1 - K_t R^{-1}) P_{t|t-1}$ where Q is the process noise covariance (1×10^{-5}) and R is the measurement noise covariance (1×10^{-3}).

4.2 Dynamic Leverage Scaling (DLS)

We dynamically scale position leverage L_t to safeguard against sudden downside shocks: $L_t = \text{clip}\left(\frac{0.15}{\sigma_{30d}}, 1.0, L_{\max}\right)$ where: * σ_{30d} is the rolling 30-day annualized return volatility. * $L_{\max}$ is set to $4.0 \times$ for equities and $1.5 \times$ for cryptocurrencies.

5. Denoised Kelly Sizing & Leverage Peak Profit Sweep

The standard Kelly Criterion defines the optimal bet size fraction f^* to maximize long-term log-utility: $f^* = \frac{p \cdot b - q}{b}$ where p is the win probability, $q = 1 - p$, and b is the win/loss ratio.

We performed a parameter grid sweep from $0.1 \times$ to $3.0 \times$ (steps of $0.05 \times$) over constant sizing multipliers on Kalman Filter Denoised (KFD) price series.

5.1 Peak Profit Sweep Results (600 Days)

The grid sweep identified a global peak profit boundary at a Kelly multiplier of **3.00x**:

Asset	Platform	Static Peak Mult	Static Peak Return	Static Peak DD	Mitigated Peak Return	Mitigated Peak DD
BTC-USD	Coinbase (Crypto)	3.00x	769.05%	10.99%	769.05%	10.99%
PLTR	Alpaca (Stock)	3.00x	17,212,448.30%	42.05%	11,305.56%	20.75%
COIN	Alpaca (Stock)	3.00x	9,119,663.42%	53.90%	2,288.43%	33.47%
AVGO	Alpaca (Stock)	3.00x	5,301,432.47%	42.31%	4,469.58%	27.64%

6. Transfer Entropy Early Warning System (TE-EWS)

To address transition lag in traditional regime-switching models (e.g. HMM), we utilize cross-asset Symbolic Transfer Entropy (STE).

6.1 Formulation

We compute the transfer entropy from a market index leader X (e.g. SPY or BTC) to a target asset Y : $STE(X \rightarrow Y) = H(Y_t | Y_{t-1}) - H(Y_t | Y_{t-1}, X_{t-\tau})$ where H represents Shannon entropy. If $STE(X \rightarrow Y)$ crosses a threshold value γ , the early-warning system triggers a defensive sizing reduction ($0.5 \times$) to protect capital before Viterbi state-transitions manifest in Y 's local volatility.

7. Limit Order Book (L2 LOB) Queue Position Forecasting

We model our fill probability $P(t)$ inside the limit order book queue to manage execution slippage: $P(t) = 1.0 - \Phi\left(\frac{Q_{\text{queue}} - \lambda \cdot t}{\sigma \sqrt{t}}\right)$ where: * Q_{queue} is the initial queue position (in shares) behind existing orders. * λ is the arrival rate of aggressive market orders crossing the spread. * σ is the order flow variance. * Φ is the standard normal CDF.

8. Academic References

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